

Q1. DIRECTIONS *for question 1:* Select the correct alternative from the given choices.

In a list of 11 numbers, seven numbers are 8, 9, 12, 5, 10, 8 and 9. Which of the following gives the least possible value of the median of all the 11 numbers?

- ☐ a) 5
- ☐ b) 8
- ☐ c) 9
- ☐ d) None of the above

Q2. DIRECTIONS *for questions 2 and 3:* Type in your answer in the input box provided below the question.

In how many ways can twelve similar balls be divided into three groups, with each group containing at most six balls?

Q3. DIRECTIONS *for questions 2 and 3:* Type in your answer in the input box provided below the question.

Akhil was multiplying two 2-digit numbers, AB and CD. He accidentally interchanged the digits of one of the numbers and the product he obtained differed from the correct product by a value d . How many of the following values can d assume?

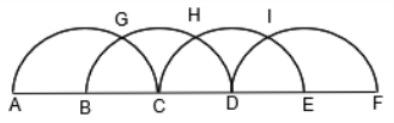
i. 1234

ii. 2345

iii. 3456

iv. 9495

Q4. DIRECTIONS for question 4: Select the correct alternative from the given choices.



A, B, C, D, E and F are evenly spaced points on a line. Four semicircles, each with radius r and centres at B, C, D and E, are drawn on the same side of the line AF as shown in the figure above. What is the length of the curve AGHIF?

- ☐ a) $\frac{5}{3} \pi r$
- ☐ b) $\frac{7}{3} \pi r$
- ☐ c) $\frac{2}{3} \pi r$
- ☐ d) None of the above

Q5. DIRECTIONS *for questions 5 and 6:* Type in your answer in the input box provided below the question.

Indian Airlines has a certain free luggage allowance for each passenger. It charges for excess luggage at a fixed rate per kg. Two passengers, Mohan and Sohan, have a total of 50 kg of luggage between them. They were charged ₹2800 and ₹1400 respectively for excess luggage. If the free luggage allowance were halved and the entire luggage belonged to one of them, the excess luggage charge would have been ₹6300. Find the weight (in kg) of Mohan's luggage.

Q6. DIRECTIONS *for questions 5 and 6:* Type in your answer in the input box provided below the question.

A pipe, P, can fill an empty tank in 12 minutes, while another pipe, Q, can empty the same tank completely in 20 minutes. Pipe P closes automatically when the water level in the tank reaches $\frac{3}{4}$ th of the height of the tank, simultaneously the pipe Q opens and operates until the water level falls to exactly $\frac{1}{4}$ th of the height of the tank, after which Q closes and simultaneously P opens. If the pipe P is opened first and this process continues indefinitely, F is the fraction of the tank that is filled after exactly one hour, find the value of 20F. (Assume that the tank is initially empty)

Q7. DIRECTIONS *for questions 7 to 11:* Select the correct alternative from the given choices.

If $a : b = 3 : 2$; $b : c = 4 : 5$; $c : d = 5 : 6$ and $d : e = 6 : 5$, find the value of $\frac{ad+ce}{bc+ae}$.

☒ a) 1.22 ✓ Your answer is correct

☐ b) 1.33

☐ c) 1.44

☐ d) 1.66

Q8. DIRECTIONS for questions 7 to 11: Select the correct alternative from the given choices.

If $\log_a 15 = x$, $\log_a 45 = y$ and $\log_a 75 = z$, which of the following would be equal to $\log_a 10125$.

☐ a) $\frac{2x + 4y + 2z}{3}$

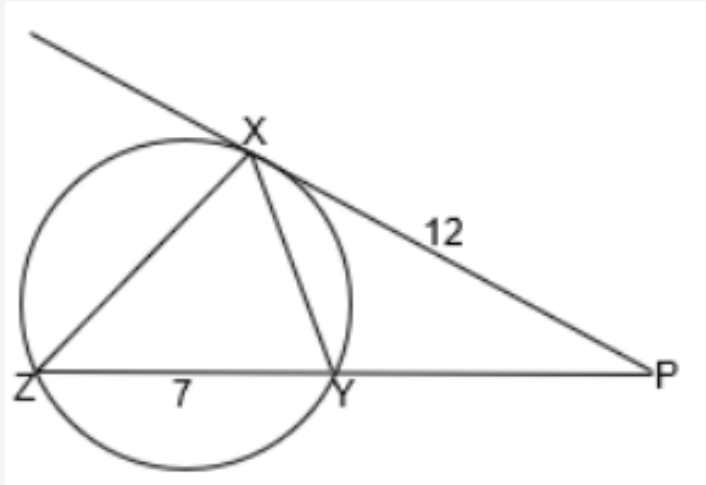
☐ b) $\frac{2x + 4y + 4z}{3}$

☐ c) $\frac{3x + 4y + z}{3}$

☐ d) $\frac{3x + 2y + z}{3}$

Q9. DIRECTIONS for questions 7 to 11: Select the correct alternative from the given choices.

In the figure below, $PX = 12$ cm, $YZ = 7$ cm and the perimeter of PXY is 27 cm. Find the perimeter of ΔPXZ .



- ☐ a) 36 cm
- ☐ b) 32 cm
- ☐ c) 34.5 cm
- ☐ d) 31.5 cm

Q10. DIRECTIONS *for questions 7 to 11:* Select the correct alternative from the given choices.

The lecturers of a certain college contributed for a charity programme. All lectures of the college are classified as either senior lecturers or junior lecturers. The senior lecturers contributed an average of ₹250 per head and the junior lecturers contributed an average of ₹100 per head. If the average contribution per lecturer was ₹160 on the whole, what percentage of the lecturers are junior lecturers?

- ☐ a) 55%
- ☒ b) 60% ✓ **Your answer is correct**
- ☐ c) 65%
- ☐ d) 80%

Q11. DIRECTIONS *for questions 7 to 11:* Select the correct alternative from the given choices.

The savings of Achyut are 30% of his earnings, while the savings of Ranjit are 150% of the savings of Achyut. The combined earnings of Achyut and Ranjit as a p

☐ a) 150%.

☐ b) 160%.

☐ c) 120%.

☐ d) 180%.

Q12. DIRECTIONS *for question 12:* Type in your answer in the input box provided below the question.

If θ is an angle (in radians) in the first quadrant of the x - y plane, and the solution of the equation

$$8^{1+|\sin\theta|+\sin^2\theta+|\sin^3\theta|+\sin^4\theta+\dots+\infty} = 4^3 \text{ is } \theta = \frac{\pi}{n}, \text{ find } n.$$

Q13. DIRECTIONS *for question 13:* Select the correct alternative from the given choices.

A cube is taken, and all its six faces are painted. Then, the cube is cut by six planes, so as to yield the maximum number of smaller cubes. If 3.5 litres of paint was consumed to paint the original cube, find the additional quantity of paint needed to paint all the unpainted faces of all the smaller cubes.

☐ a) 3.5 litres

☐ b) 11.5 litres

☐ c) 10.5 litres

☐ d) 7 litres

DIRECTIONS for questions 14 and 15: Answer the questions on the basis of the information given below.

A function $S(n)$ takes a natural number n as its argument and gives the output as the sum of n and the number formed by reversing the digits of n .

Another function $D(n)$ takes the natural number n as its argument and gives the output as the difference between n and the number formed by reversing the digits of n .

Q14. DIRECTIONS for question 14: Select the correct alternative from the given choices.

If k is a three-digit natural number and $D[D(D(k))] = 0$, but $D(D(k)) \neq 0$, which of the following cannot be the value of k ?

☐ a) 762

☐ b) 166

☐ c) 495

☐ d) 208

Q15. DIRECTIONS *for question 15:* Type in your answer in the input box provided below the question.

If k is a three-digit natural number with its digits from one end to the other in either ascending order or descending order, from one end to the other, how many values can $S(D(k))$ take?

Q16. DIRECTIONS *for question 16:* Type in your answer in the input box provided below the question.

In a box there are 90 slips, numbered from 1 to 90. Ritu picks up four slips at random, one after the other, without replacement. If P is the probability that the numbers on the slips, in the order she picks them up, are in ascending order, find the value of $120P$?

Q17. DIRECTIONS *for questions 17 and 18:* Select the correct alternative from the given choices.

Chandrakanth took a loan of ₹50,000, at an interest rate of 10% per annum, interest being compounded annually. If he repaid ₹10,000 at the end of the first year and an equal amount at the end of each of the second, third and the fourth years, find the amount he has to pay at the end of the fifth year to repay the entire loan.

- ☐ a) ₹20,050
- ☐ b) ₹22,275.5
- ☐ c) ₹26,795
- ☐ d) ₹29,474.5

Q18. DIRECTIONS *for questions 17 and 18:* Select the correct alternative from the given choices.

It has been planned to connect the five cities Delhi, Mumbai, Chennai, Kolkata and Bangalore through a network of super-expressways, with certain conditions. Each of the five cities should be connected to at least three other cities. Also, Delhi, being the national capital, should be connected to all the other cities. What is the minimum number of super-expressways required for this plan?

☐ a) 8

☐ b) 10

☐ c) 7

☐ d) 9

Q19. DIRECTIONS *for question 19:* Type in your answer in the input box provided below the question.

A number N^2 , where N is a natural number, is such that exactly three of its factors are less than N . What is the number of factors of N^3 ?

Q20. DIRECTIONS *for question 20:* Select the correct alternative from the given choices.

During Diwali season, Jaideep a shop owner marked the price of an article 50% above the cost price and then offered a discount of 40%. Find the profit or loss percent made by Jaideep in selling the article.

- ☐ a) 10% profit
- ☒ b) 10% loss ✓ **Your answer is correct**
- ☐ c) 30% profit
- ☐ d) 30% loss

DIRECTIONS for questions 21 and 22: Answer the questions on the basis of the information given below.

There are 360 poles, fixed symmetrically, all around a circular track. Four runners A, B, C and D, start simultaneously from the first pole and run, all in the same direction, around the track at speeds of a , b , c and d respectively.

Q21. DIRECTIONS for questions 21 and 22: Select the correct alternative from the given choices.

For which of the following sets of speeds (in m/s) will two or more runners meet at a point *between* two consecutive poles?

- ☐ a) 1, 4, 6, 10
- ☐ b) 2, 3, 6, 9
- ☐ c) 1, 2, 3, 7
- ☐ d) More than one of the above

DIRECTIONS for questions 21 and 22: Answer the questions on the basis of the information given below.

There are 360 poles, fixed symmetrically, all around a circular track. Four runners A, B, C and D, start simultaneously from the first pole and run, all in the same direction, around the track at speeds of a , b , c and d respectively.

Q22. DIRECTIONS for questions 21 and 22: Select the correct alternative from the given choices.

If all the four runners do not run in the same direction, then for which of the following sets of speeds (in m/s) is it *not* possible for two or more runners to meet at a point on the track between two consecutive poles?

- ☐ a) 1, 4, 6, 10
- ☐ b) 2, 3, 6, 9
- ☐ c) 1, 2, 3, 7
- ☐ d) More than one of the above